

ITA06 - Machine Learning

Exp 12 :Implement Iris Flower Classification using KNN

Aim

To implement the **K-Nearest Neighbors (KNN)** algorithm for classification and predict the class label of new samples using different datasets.

Software Requirements

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn

Algorithm

1. Import the required libraries.
2. Create the dataset using Pandas DataFrame.
3. Encode categorical target values if necessary.
4. Split the data into features (X) and target (y).
5. Train the KNN Classifier using the dataset.
6. Provide a new sample for prediction.
7. Predict the class label.
8. Display the result.

Procedure

1. Import the Iris dataset and KNN classifier.
2. Load the Iris dataset.
3. Train the KNN model using the training data.
4. Predict the flower class.
5. Display the predicted result.

Dataset 1: Fruit Classification

Weight (g)	Diameter (cm)	Sweetness (%)	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
145	7.3	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6.0	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

Target Variable

Fruit: Apple / Orange

New Sample

Weight	Diameter	Sweetness
152	7.5	90

Dataset 2: Student Placement Classification

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed

CGPA Aptitude Score Communication Score Placement

6.8	68	65	Not Placed
9.0	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

Target Variable

Placement: Placed / Not Placed

New Sample

CGPA Aptitude Score Communication Score

8.7	86	84
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Dataset 3: Disease Diagnosis

Temperature (°C) Heart Rate (bpm) Oxygen Level (%) Disease

39.0	110	94	Positive
38.5	108	95	Positive
37.0	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative

Temperature (°C) Heart Rate (bpm) Oxygen Level (%) Disease

38.9 111 93 Positive

Target Variable

Disease: Positive / Negative

New Sample

Temperature Heart Rate Oxygen Level

38.7 109 94

Dataset 4: Employee Promotion Prediction

Experience (Years) Performance Score Training Hours Promotion

10	95	50	Promoted
8	90	45	Promoted
7	88	40	Promoted
3	65	20	Not Promoted
2	60	18	Not Promoted
9	93	48	Promoted
4	70	25	Not Promoted
8	89	42	Promoted
2	58	15	Not Promoted
6	85	38	Promoted

Target Variable

Promotion: Promoted / Not Promoted

New Sample

Experience Performance Score Training Hours

7 90 44

Dataset 5: Vehicle Classification

Engine Capacity (cc) Weight (kg) Mileage (km/L) Vehicle Type

100	95	70	Bike
125	110	65	Bike
150	125	55	Bike
800	850	22	Car
1000	950	20	Car
1200	1050	18	Car
110	100	68	Bike
900	900	21	Car
140	120	58	Bike
1300	1100	17	Car

Target Variable

Vehicle Type: Bike / Car

New Sample

Engine Capacity Weight Mileage

135 118 60

[1]
✓ 2s

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

X = [
    [150,7.5,90],[160,7.8,88],[145,7.3,91],
    [120,5.5,70],[125,5.8,72],[130,6.0,74],
    [155,7.6,89],[118,5.4,69],[148,7.4,92],[122,5.7,71]
]

y = ["Apple","Apple","Apple","Orange","Orange",
     "Orange","Apple","Orange","Apple","Orange"]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)

sample = [[152,7.5,90]]
print("Predicted Fruit:", model.predict(sample)[0])
```

✓ ... Predicted Fruit: Apple

1s

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

X = [
    [9.2,90,88],[8.8,85,82],[8.5,80,79],
    [7.2,72,70],[6.8,68,65],[9.0,92,90],
    [7.5,75,74],[8.6,84,81],[6.5,60,58],[8.9,88,86]
]

y = ["Placed","Placed","Placed","Not Placed","Not Placed",
     "Placed","Not Placed","Placed","Not Placed","Placed"]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)

sample = [[8.7,86,84]]
print("Predicted Placement:", model.predict(sample)[0])
```

Predicted Placement: Placed

3]
✓ 0s

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

X = [
    [39.0,110,94],[38.5,108,95],[37.0,78,99],
    [39.2,115,93],[36.8,75,98],[38.8,112,94],
    [37.2,80,98],[39.1,114,92],[36.7,74,99],[38.9,111,93]
]

y = ["Positive","Positive","Negative","Positive","Negative",
     "Positive","Negative","Positive","Negative","Positive"]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)

sample = [[38.7,109,94]]
print("Predicted Disease:", model.predict(sample)[0])
```

✓ Predicted Disease: Positive

```
1  ▶ import pandas as pd
2  from sklearn.neighbors import KNeighborsClassifier

X = [
    [10,95,50],[8,90,45],[7,88,40],
    [3,65,20],[2,60,18],[9,93,48],
    [4,70,25],[8,89,42],[2,58,15],[6,85,38]
]

y = ["Promoted","Promoted","Promoted","Not Promoted",
     "Not Promoted","Promoted","Not Promoted",
     "Promoted","Not Promoted","Promoted"]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)

sample = [[7,90,44]]
print("Predicted Promotion:", model.predict(sample)[0])

... Predicted Promotion: Promoted
```

```
1  ▶ import pandas as pd
2  from sklearn.neighbors import KNeighborsClassifier

X = [
    [100,95,70],[125,110,65],[150,125,55],
    [800,850,22],[1000,950,20],[1200,1050,18],
    [110,100,68],[900,900,21],[140,120,58],[1300,1100,17]
]

y = ["Bike","Bike","Bike","Car","Car",
     "Car","Bike","Car","Bike","Car"]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)

sample = [[135,118,60]]
print("Predicted Vehicle Type:", model.predict(sample)[0])

... Predicted Vehicle Type: Bike
```

Result

Thus, the K-Nearest Neighbors (KNN) Classification algorithm was successfully implemented on multiple datasets. The model accurately classified new samples into their respective categories such as Apple/Orange, Placed/Not Placed, Positive/Negative, Promoted/Not Promoted, and Bike/Car based on the nearest neighboring data points.